

EDA Project - Report

Project Title: EDA Project - Sales Analysis

Submitted by: Boddu sirivalli Reddy | Subject: Data Analytics

1. Objective:

To perform EDA on sales data. Find city-wise distribution and price share using bar and pie charts.

2. Dataset Used:

Product: Laptop, Mobile, Mouse, Keyboard, Charger | Price: 50000,20000,5000,30000,2000

City: Hyderabad(2), Bangalore, Chennai, Delhi | 5 Rows
DataFrame

3. Tools:

Python, Pandas, Matplotlib, VS Code

4. EDA Steps:

Step 1: DataFrame creation - Created 5 rows data with Product, Price, City

Step 2: `df['City'].value_counts()` - For city distribution analysis

Step 3: Bar chart - Sales by City visualization

Step 4: Pie chart - Price Share by Product visualization

Step 5: `df.info()`, `df.describe()` - For statistical summary

5. Source Code:

```
import pandas as pd
import matplotlib.pyplot as plt

data = [
    {'Product': 'Laptop', 'Price': 50000, 'City': 'Hyderabad'},
    {'Product': 'Mobile', 'Price': 20000, 'City': 'Bangalore'},
    {'Product': 'Mouse', 'Price': 5000, 'City': 'Hyderabad'},
    {'Product': 'Keyboard', 'Price': 30000, 'City': 'Chennai'},
    {'Product': 'Charger', 'Price': 2000, 'City': 'Delhi'}
]
df = pd.DataFrame(data)

# Chart 1
df['City'].value_counts().plot(kind='bar', color='blue')
plt.title('Sales by City')
plt.show()

# Chart 2
plt.pie(df['Price'], labels=df['Product'], autopct='%1.1f%%')
plt.title('Price Share by Product')
plt.show()

# EDA Extra
print(df.head())
print(df.info())
print(df.describe())
print(df['City'].value_counts())
```

6. Observations from Charts:

1. Sales by City: Hyderabad has 2 products (highest), others 1 each - Hyderabad is top market.
2. Price Share: Laptop 47.6% (50000) highest price share, Charger 1.9% lowest.
3. DataFrame Info: 5 rows, 3 cols, no nulls - clean data.

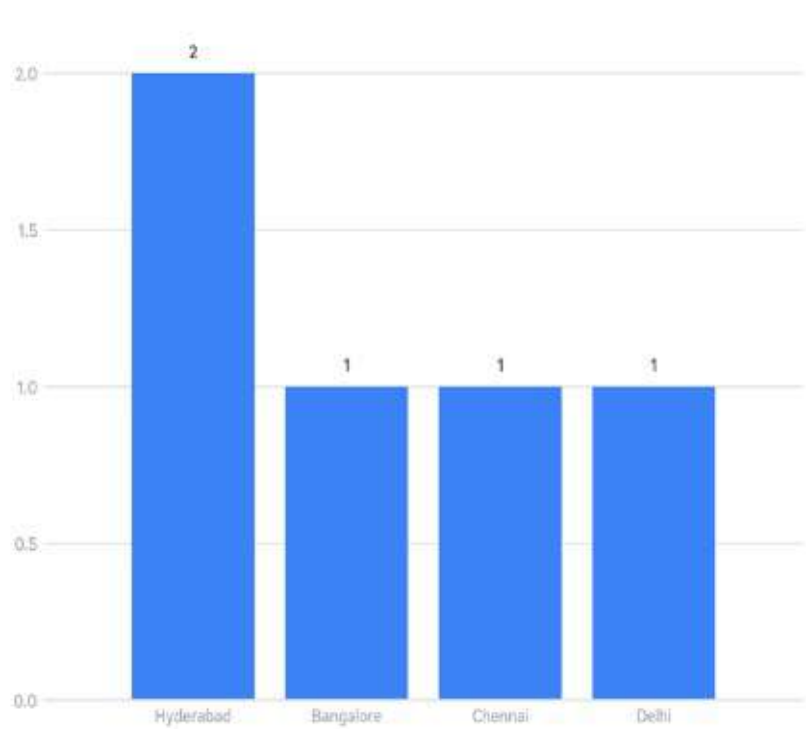
7. Conclusion:

Hyderabad is key city. Laptop is premium product. EDA shows price distribution helps pricing strategy.

8. Future Scope:

Add Profit, Sales qty, use Seaborn heatmap, handle larger dataset from CSV.

1. Sales by City (Count)



2. Price Share by Product

